

Fully Online Retrieval Adaptation

Causal problem formulation and proposed ridge adaptor

Thesis project

Standalone method note

Abstract

This note isolates the scientific formulation from implementation and cluster operations. It is intended to be readable on its own and reusable when drafting a paper introduction or method section.

1 Online forecasting task

For user u and query date s , the frozen backbone observes $X_{u,s} = x_u(s - L + 1:s)$ and predicts $Y_{u,s} = x_u(s + 1:s + H)$. Every object used at s must be measurable before the prediction is made.

2 Causal datastore and fitting sets

A candidate window ending at r is eligible only if $r + H \leq s$. The datastore is therefore

$$\mathcal{D}_s = \{(X_{v,r}, Y_{v,r}) : r + H \leq s, (v, r) \text{ satisfies the configured scope and capacity}\}.$$

Retrieval returns the first K elements of one ranking computed up to $K_{\max}$. Fitting dates are selected independently:

$$\mathcal{F}_{u,s} = \{r_1, \dots, r_{N_{\text{fit}}} : r_i + H \leq s\},$$

with either one set per user or one all-user set per date.

3 Proposed adaptor

For each fitting example, let $\phi_{u,r,K}$ contain the declared vanilla, retrieval-context, covariance, and retrieved-target signals. Shared ridge fits

$$\hat{\beta}_{s,\alpha,K} = \arg \min_{\beta} \sum_{(u,r) \in \mathcal{F}_s} \|Y_{u,r} - \phi_{u,r,K}^\top \beta\|_2^2 + \alpha \|\beta\|_2^2.$$

Features are divided by their fitting-set root mean square before regularization. The newest fitting subset selects (α, K) , ties prefer smaller K and stronger regularization, and the selected pair is refit on the complete causal fitting set.

4 Prediction and alternatives

The online prediction is

$$\hat{Y}_{u,s} = \phi_{u,s,K_s}^\top \hat{\beta}_s.$$

Matched alternatives include a simplex-constrained mixture, a causal soft Bayes gate, a parameter-free covariate-informed prediction, vanilla inference, and source-adapted TS-RAG on its native fixed protocol.

5 Claims tested

The study asks whether adaptation improves average and worst-user error, whether gains survive fixed/rolling and scope controls, and whether the accuracy improvement justifies extraction, fitting, and cold-batch cost. Methods are compared only on identical evaluation dates.